Electric Power Steering Control Module Specification

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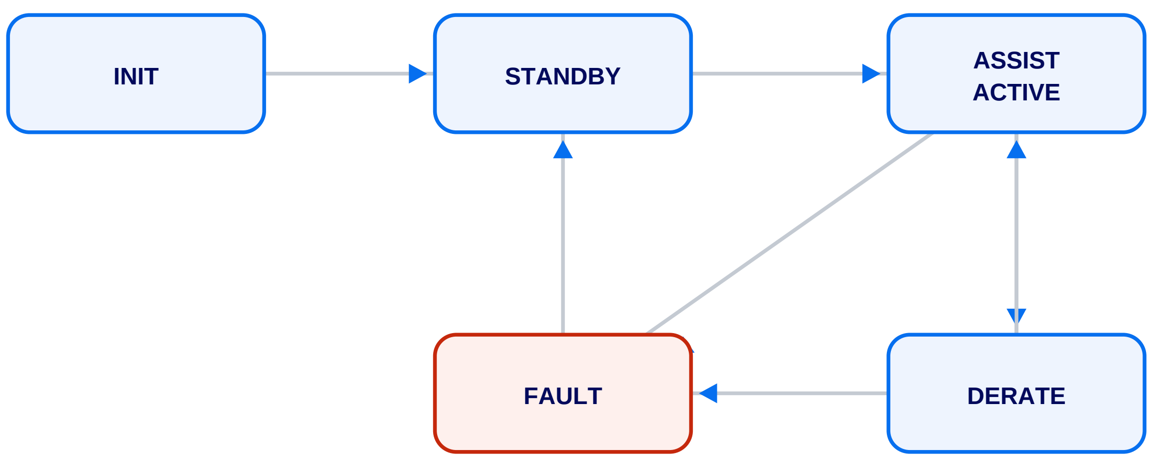
This specification defines the control logic, diagnostic coverage, communication signals, and calibration parameters for the Electric Power Steering (EPS) control module. The assist state machine governs transitions between standby, active assist, thermal derate, and fault handling. All diagnostic trouble codes are debounced per the table below.

1. Assist State Machine

The module transitions through the following states. Note the multiple paths into FAULT and the recovery path back to STANDBY — reconstructing this from prose alone is error-prone.



2. Diagnostic Trouble Codes (DTC Matrix)

DTC Code	Description	Fault Action	MIL	Debounce (ms)
C1511	Torque Sensor Signal Implausible	Disable assist, latch	Yes	200
C1513	Motor Phase U Open Circuit	Derate to 40%	Yes	500
C1515	Motor Over-Temperature	Thermal derate	No	1000
C1521	Steering Angle Sensor No Comms	Inhibit assist	Yes	300
U0126	Lost Comms with SAS Module	Use estimated angle	No	400
P0AA6	HV System Isolation Fault	Open contactors	Yes	100
C1A00	ECU Internal Watchdog Reset	Safe-state, reset	Yes	50

3. CAN Signal Definitions

Signal Name	Message ID	Start Bit	Length	Scale	Unit	Cycle (ms)
EPS_AssistTorque	0x1A4	0	12	0.1	Nm	10
EPS_MotorCurrent	0x1A4	12	10	0.05	A	10
EPS_SteeringAngle	0x0C8	0	16	0.0625	deg	10
EPS_AngleRate	0x0C8	16	12	0.25	deg/s	10
EPS_ModuleState	0x2F0	0	4	1	enum	100
EPS_TempMotor	0x2F0	8	8	1	degC	100
EPS_FaultActive	0x2F0	16	1	1	bool	100

4. Calibration Parameters

Parameter ID	Name	Min	Max	Default	Unit
K_ASSIST_GAIN	Base Assist Gain	0.5	3.0	1.8	Nm/Nm
K_RETURN_GAIN	Active Return Gain	0.0	1.5	0.6	-
K_DAMP_HI	High-Speed Damping	0.0	2.0	0.9	Nm/(deg/s)
T_DERATE_ON	Thermal Derate Onset	90	140	115	degC
T_DERATE_OFF	Thermal Derate Recovery	70	120	95	degC
V_ASSIST_MAX	Max Assist Speed	60	250	200	kph